

Conceptual Data Model

- Entities:
- Customer
- Represents an individual customer with identity and contact details
- Order
- Represents a transaction or purchase made by a customer
- OrderSummary
- Represents an integrated view of customer and order information, maintaining the full change history (SCD Type 2)
- CustomerAggregateSpend
- Represents daily aggregated total spend per customer
- Key Relationships:
- Customer "places" one or more Orders
- OrderSummary is derived by joining Customer and Order on CustId (with SCD Type 2 tracking)
- CustomerAggregateSpend aggregates OrderSummary by Name and Date
- Business Definitions:
- Customer: The party purchasing goods or services
- Order: A transactional record of a purchase or sale
- OrderSummary: A historical fact table containing both customer and order details
- CustomerAggregateSpend: The sum of all orders (by customer) per day

Logical Data Model

- Entities, Attributes, Keys, Relationships:
- Customer
- CustId (Primary Key)
- Name
- EmailId
- Region
- Order
- OrderId (Primary Key)
- ItemName
- PricePerUnit
- Qty
- Date

- CustId (Foreign Key → Customer)
- OrderSummary
- CustId (Foreign Key → Customer)
- Name
- EmailId
- Region
- OrderId
- ItemName
- PricePerUnit
- Qty
- Date
- (Implied SCD Type 2 fields: StartDate, EndDate, Active)
- CustomerAggregateSpend
- Name
- TotalAmount
- Date
- Cardinality and Relationships:
 - Customer 1 — * Order (one-to-many)
 - Order * — 1 Customer (many-to-one)
 - OrderSummary 1 — 1 Customer + 1 Order (composite join)
 - CustomerAggregateSpend * — 1 Name, 1 Date (aggregated facts)
- Normalization Notes:
 - Customer and Order are 3NF (no repeating groups, single facts per row)
 - OrderSummary denormalizes for historical tracking with SCD Type 2 (duplication expected)
 - CustomerAggregateSpend is an aggregate fact (denormalized for performance)
- Business Rules per Entity:
 - Customer: No duplicates, all fields cleansed for literal or "Null"/Null values
 - Order: Must be linked to Customer (CustId), no duplicates/"Null" values
 - OrderSummary: SCD Type 2—maintain full-history, status, active/inactive records
 - CustomerAggregateSpend: Summed TotalAmount grouped by Name and Date

Physical Data Model

- Physical Structures (Delta Tables):

- customer
- CustId:
- Name:
- EmailId:
- Region:
- order
- OrderId:
- ItemName:
- PricePerUnit:
- Qty:
- Date:
- CustId:
- ordersummary
- CustId:
- Name:
- EmailId:
- Region:
- OrderId:
- ItemName:
- PricePerUnit:
- Qty:
- Date:
- StartDate:
- EndDate:
- Active:
- customeraggregatespend
- Name:
- TotalAmount:
- Date:
- Indexing Strategy:
- Primary keys and Foreign keys implied, with suggested indexes on CustId, Date in transactional and aggregate tables
- Partitioning, Clustering, Storage:

- Partition by Date for order, ordersummary, and customeraggregatespend for efficient time-based queries
- Cluster by CustId in order, ordersummary tables for customer-centric workloads
- Constraints:
 - Primary Key: CustId (customer), OrderId (order)
 - Foreign Key: CustId (order, ordersummary)
 - (Implied) Not Null constraints on key columns
 - Uniqueness: OrderId (order), CustId + Date (ordersummary for SCD version records)
- Performance Optimization Notes:
 - Use Delta tables for ACID compliance and efficient change-data capture
 - Store as Parquet or Delta format for columnar performance

Naming Conventions

- Table names: lower case, underscores between words (customer, order, ordersummary, customeraggregatespend)
- Column names: PascalCase (CustId, OrderId, PricePerUnit, TotalAmount, etc.)

Data Dictionary

Entity	Attribute	Type	Description
customer	CustId	<TBD>	Unique customer identifier
customer	Name	<TBD>	Customer name
customer	EmailId	<TBD>	Customer email address
customer	Region	<TBD>	Customer region
order	OrderId	<TBD>	Unique order identifier
order	ItemName	<TBD>	Item name
order	PricePerUnit	<TBD>	Price per single unit
order	Qty	<TBD>	Quantity ordered
order	Date	<TBD>	Order date
order	CustId	<TBD>	Reference to customer
ordersummary	CustId	<TBD>	Customer identifier
ordersummary	Name	<TBD>	Customer name
ordersummary	EmailId	<TBD>	Customer email address
ordersummary	Region	<TBD>	Customer region
ordersummary	OrderId	<TBD>	Order identifier
ordersummary	ItemName	<TBD>	Item purchased

ordersummary	PricePerUnit	<TBD>	Price per unit for the item
ordersummary	Qty	<TBD>	Quantity ordered
ordersummary	Date	<TBD>	Order date
ordersummary	StartDate	<TBD>	SCD Type 2 start date
ordersummary	EndDate	<TBD>	SCD Type 2 end date
ordersummary	Active	<TBD>	Current status flag (active/inactive)
customeraggregatespend	Name	<TBD>	Customer name
customeraggregatespend	TotalAmount	<TBD>	Total amount spent
customeraggregatespend	Date	<TBD>	Aggregation date

Text-Based ERD Description

[Customer]

*CustId

Name

EmailId

Region

[Order]

*OrderId

ItemName

PricePerUnit

Qty

Date

CustId

[OrderSummary]

*CustId

Name

EmailId

Region

OrderId

ItemName

PricePerUnit

Qty

Date

StartDate

EndDate

Active

[CustomerAggregateSpend]

*Name

TotalAmount

Date

Customer *--* Order

OrderSummary *--1 Customer

OrderSummary *--1 Order

CustomerAggregateSpend *--* OrderSummary